

Uncertainty-Driven Knowledge Distillation for Language Model Compression

Tianyu Huang[✉], Weisheng Dong[✉], *Member, IEEE*, Fangfang Wu[✉], Xin Li[✉], *Fellow, IEEE*,
and Guangming Shi[✉], *Fellow, IEEE*

Abstract—Despite the remarkable performance on various Natural Language Processing (NLP) tasks, the parametric complexity of pretrained language models has remained a major obstacle due to limited computational resources in many practical applications. Techniques such as knowledge distillation, network pruning, and quantization have been developed for language model compression. However, it has remained challenging to achieve an optimal tradeoff between model size and inference accuracy. To address this issue, we propose a novel and efficient uncertainty-driven knowledge distillation compression method for transformer-based pretrained language models. Specifically, we design a method of parameter retention and feedforward network parameter distillation to compress N-stacked transformer modules into one module in the fine-tuning stage. A key innovation of our approach is to add the uncertainty estimation module (UEM) into the student network such that it can guide the student network’s feature reconstruction in the latent space (similar to the teacher’s). Across multiple datasets in the natural language inference tasks of GLUE, we have achieved more than 95% accuracy of the original BERT, while only using about 50% of the parameters.

Index Terms—Knowledge distillation, model compression, uncertainty estimate.

I. INTRODUCTION

TRANSFER Learning has emerged in the field of natural language processing (NLP) as a common tool for leveraging data from additional domains to train a model with good generalization properties [17], [33], [34]. Pre-training on large-scale corpus and fine-tuning on downstream tasks has become a standard problem-solving paradigm in both NLP and vision [1], [22]. The complexity and depth of the network continue

to grow and surpass previous architectures - e.g., large-scale networks improve the performance of the network by stacking N general modules [24]. Transformer-based pretraining models benefit from their over-parameterization nature, achieving the best performance in the General Language Understanding Evaluation (GLUE). However, these models usually contain millions or even billions of parameters, making them impractical for real-world applications. Memory consumption and high latency greatly hinder the application of these models in both industrial and academic research.

As the scale of pretrained language models (PLMs) continues to expand, model compression has emerged as a promising remedy. Existing main research directions can be divided into quantization [25], [32], pruning [10], [30], [31], and knowledge distillation [13], [16], [18], [39], [48]. Recently, the idea of dynamic networks [45], [46], [47] has arisen. The knowledge distillation scheme is particularly appealing for the task of language model compression. The main idea of knowledge distillation is to use large-scale networks as a teacher network to provide soft labels for a student network to learn. The process allows the student network to acquire more guidance and knowledge from the teacher network while learning a specific task. Since knowledge distillation does not have specific restrictions on the compressed network structure, most of the knowledge distillation methods [18], [20] pay less attention to the uncertainty of the network output distribution of teacher and student networks.

Generally speaking, the generation of large-scale pretrained language models is usually divided into two stages: pretraining and fine-tuning. Applying knowledge distillation constraints in the pretraining stage [6] can achieve better performance because the pretraining stage contains most of the semantic information learned from the training data. A previous study [21] has shown that knowledge distillation in the fine-tuning stage often leads to inferior performance when compared with the pretraining alternative. However, distillation in the pretraining stage requires prohibitive computing resources, which are often beyond the reach of common users (e.g., the need to access a cloud computing platform with high computation density). Since we want to reduce the computational burden by skipping the pretraining stage, how to exploit *uncertainty modeling* to facilitate the task of knowledge distillation in the fine-tuning stage serves as a primary motivation for this work.

In this article, we propose a novel knowledge distillation method for transformer-based models that compresses N

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Tianyu Huang, Weisheng Dong, and Guangming Shi are with the School of Artificial Intelligence, Xidian University, Xi’an 710071, China (e-mail: 19171213910@stu.xidian.edu.cn; wsdong@mail.xidian.edu.cn; gmshi@xidian.edu.cn).

Fangfang Wu is with the School of Computer Science and Technology, Xidian University, Xi’an 710071, China (e-mail: wufangfang@xidian.edu.cn).

Xin Li is with the Department of Computer Science, University at Albany, Albany, NY 12222 USA (e-mail: xli48@albany.edu).

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stacked modules into one module. We first train the pretrained BERT language model on specific tasks until reaching convergence. Then we use the fine-tuned BERT network by copying the parameters of the odd-numbered layers as the initialized parameters for the student network. Finally, we freeze the student network weights of the attention structure and distill the weight parameters of the feedforward network while fine-tuning. By retaining the parameters of odd-numbered layers and the distillation network parameters of even-numbered layers, we strive to preserve the representational capabilities of the original pretrained model as much as possible. A key new insight brought by this work is *uncertainty-driven knowledge distillation* - by adding a new branch of network for modeling the uncertainty, we show how to significantly improve the performance of knowledge distillation on language model compression. Uncertainty modeling allows us to solve the problem of feature distillation more efficiently by fine-tuning instead of pretraining. The experimental results of ablation studies have shown that the uncertainty estimate is beneficial to knowledge distillation. The main contributions of this work are listed as follows:

- We propose a new transformer-based knowledge distillation method. Through parity replication and distillation, the training of network parameters is achieved in the fine-tuning stage, which greatly alleviates the computational burden. We show that N stacked network structures can be compressed into one structure with the negligible sacrifice of accuracy performance;
- We propose to design a new branch of the network for estimating the uncertainty (variance) of the feature distribution, which makes the student model imitate the process of feature learning by the teacher model. The uncertainty estimation module (UEM) estimates the uncertainty of the feature distribution, which measures the difficulty of learning different features in the loss function;
- The experimental results show that our student BERT network is $1.92\times$ faster and $2.2\times$ smaller than BERT-based while achieving competitive results with well-known benchmarks and achieving more than 95% accuracy of the teacher model.

II. RELATE WORK AND MOTIVATION

A. Knowledge Distillation

The purpose of knowledge distillation is to compress a large network model or an ensemble of good-performing models into a single compact model. Early pioneering work [4] is proposed by driving the student model to imitate the soft output of the teacher model. Subsequent research on knowledge distillation [6], [18], [20], [40], [41] improves the performance from several perspectives. For example, Fitnets [16] adds the intermediate representation of the teacher network to the knowledge distillation scheme. TinyBERT [6] is specially designed for knowledge distillation of transformer-based models, proposes the framework of two-stage learning, and applies attention parameter matrix distillation loss and hidden layer loss in both the pre-training and fine-tuning stages. PKD method [20] patiently learns from multiple intermediate layers of the teacher model for incremental knowledge

extraction, avoiding the over-fitting phenomenon of traditional distillation methods when only distilling the last layer of the teacher network. MobileBERT [35] reduce the data dimension of each layer by adding bottleneck layers, add a linear layer between transformer module, which greatly reduces the number of network parameters. DistilledBERT [15] uses the original BERT model as the teacher model, then conducts knowledge distillation experiments to single layer Bi-LSTM neural network on specific downstream tasks, the resulting student model has a huge speed increase while having almost the same ability as ELMo [14] in terms of performance effects.

B. Uncertainty Estimation

The prediction results given by deep learning models are not always reliable. Several studies [5], [12], [36], [43], [44] have shown that uncertainty estimation can improve the generalization ability of neural networks. Two types of uncertainty models have been studied in the literature. Aleatoric (data uncertainty) is caused by the inherent noise in the observation data, and usually, this kind of uncertainty cannot be eliminated. Epistemic (model uncertainty) is related to the model and is often caused by incomplete model training. Computer vision fields such as face recognition [19] and target detection [8] have received extensive attention. Recently, there has been corresponding uncertainty research on knowledge distillation [26]. If the deep learning model can give a higher uncertainty for wrong predictions and a lower uncertainty for correct predictions, it is possible to judge the credibility of a prediction result, and then make good use of the correctly predicted, avoiding the effects of wrong predictions. Due to the huge parameters which the pre-train language model has, to achieve a higher compression rate, the number of parameters in the student network will be much less than in the teacher network, and it is often difficult to carry all the reasoning performance of the teacher network. Therefore, how to make good use of uncertainty learning to guide knowledge distillation will be important.

In this article, we output a judgment on the credibility of the result through the uncertainty estimate model (UEM) and assign different weights to the knowledge transferred in the knowledge distillation process using this strategy.

C. Lightweight Pre-Train Language Model

Pre-trained language models have been widely used in the field of natural language processing [38]. Pre-trained networks such as ELMo [14], BERT [1], GPT [15], and XLNet [37] continue to appear and refresh the state of the art on the benchmark. Generally, pretrained language models have large parameters and deep network structures. Taking $BERT_{Base}$ as an example, $BERT_{Base}$ has 12 layers of transformer modules and 110 million parameters. Pre-training is performed on the large-scale unlabeled corpus by the mask language task and next sentence prediction task, then fine-tuning the parameters on the small-scale labeled specific tasks to obtain a network model suitable for downstream tasks. This pretraining idea exploits the massive unlabeled data in the natural language field and greatly improves the performance of the language model.

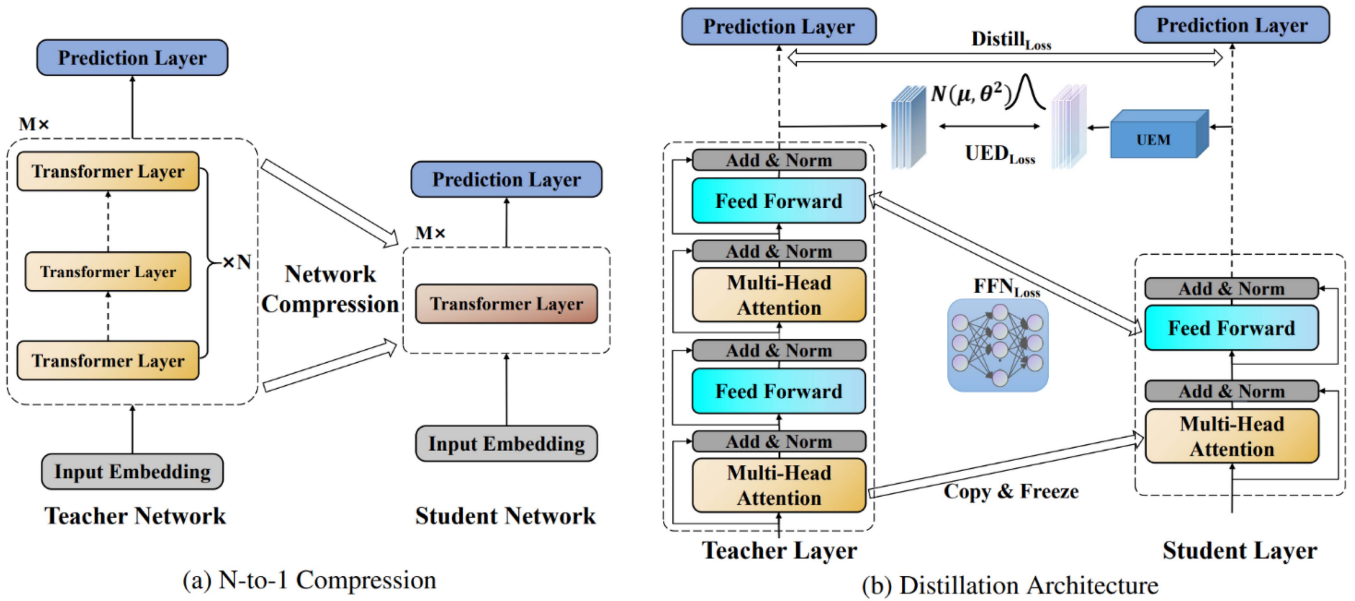


Fig. 1. Workflow of our model compression scheme. (a) N-to-1 compression compresses the stacked N consecutive layers of the transformer module into one transformer module. (b) Uncertainty-driven distillation architecture (when $N = 2$) includes copying and freezing the parameters of the odd-numbered multi-head attention (MHA) layer as well as imposing MSE constraints on the feed-forward network (FFN) of the even-numbered layer. We propose an FFN loss function and uncertainty estimate module (UEM) to improve knowledge distillation in pretrained language model compression.

Recently, the computer vision community [3] has also begun to leverage a similar mask method. However, the pretraining stage often requires a prohibitive computational burden due to the distillation of traditional knowledge, which limits its use in practice.

D. Motivation

In this article, we study the model compression problem on $BERT_{Base}$. Our method avoids the knowledge distillation process in the pretraining stage and achieves the objective of language model compression with a small training cost. Meanwhile, our method applies to other transformer-based pretrained language models. Through studying the compression problem of large-scale language models, we have found that most existing methods focus on the balance between the compression ratio and accuracy performance of the student network, but pay less attention to the computational cost during the compression process. Redirecting attention to the retention of teacher network performance by uncertainty modeling can effectively guide the training process of the student network, especially when there is a large gap in network performance between teachers (e.g., the pretrained language models) and students. Based on this observation, we copy and freeze the weight parameters of the teacher network to preserve the capability of the original network and reduce the training cost by retraining only the partial structure during the distillation/fine-tuning process.

III. METHOD

In this section, we will introduce the specific method of our approach to BERT-based model compression. We introduce the

details of the distillation method including weight replication, freezing, retraining details, and uncertainty estimation method.

Traditional knowledge distillation methods compress a large teacher network $f(x; \Theta^t)$ into a compact student network $g(x; \Theta^s)$ by defining a target L_{KD} loss. The objective is to strike the optimal tradeoff between the model cost as measured by the size of Θ^s (the smaller, the better) and the accuracy performance (the higher, the better).

A. N-to-1 Compression

Generally speaking, model compression attempts to compress a deep network into a compact network while retaining accuracy as much as possible. A popular strategy is to compress N stacked modules into one module by some network compression method, which forms a student network by distilling the compressed network modules [24], [42]. Such formulation translates the original network compression into module compression and knowledge distillation through the idea of divide-and-conquer. Along this line of reasoning, we propose a novel model compression approach to endow one transformer module with N transformer modules' capability for knowledge distillation, as shown in Fig. 1.

To facilitate the discussion, we introduce the compression strategy for $N = 2$; but our design can be easily generalized to the case of $N > 2$ (e.g., we can simply skip the middle repeating module and only retain the first multi-head attention module and the last feed-forward network parameter distillation). The teacher network $f(x; \Theta^t)$ can be defined as a deep bidirectional encoder, such as BERT, while the student network $g(x; \Theta^s)$ is a lightweight network with a reduced number of

parameters. Specifically, we take $BERT_{base}$, a 12-layer transformer model, to introduce our compression scheme, and aim at compressing $BERT_{12}$ into a smaller network of $BERT_6$ with six layers. A similar problem formulation was also used in [20] and [24].

First, we finetune the teacher network on downstream tasks and its parameters need to be learned as follows:

$$\Theta^t = \arg \min_{\Theta} \sum_{i \in [N]} L_{CE}^t(x_i, y_i; \Theta) \quad (1)$$

where the superscript t denotes that the parameter corresponds to the teacher model, (x_i, y_i) is the training pair, $[N]$ indicates the training sample set, L_{CE}^t represents the Cross-Entropy loss, and Θ denotes the parameter set of $BERT_{base}$.

Then the probability distribution of the teacher network output can be written by:

$$P^t(y_i|x_i) = softmax(f(x_i; \Theta^t)/T) \quad (2)$$

where $P^t(y_i|x_i)$ represents the probability distribution of the teacher network output; and the smoothness of the knowledge distillation can be controlled by adjusting the temperature parameter T .

Traditional knowledge distillation methods [4] often use the soft label distance between a student and a teacher as the loss penalty term:

$$L_{distill} = - \sum_{i \in [N]} \sum_{c \in C} [P^t(y_i = c|x_i; \Theta^t) \cdot \log P^s(y_i = c|x_i; \Theta^s)] \quad (3)$$

where c represents the category of the classification task, and C represents the set of class labels. The general knowledge distillation method of the language model retrains a student network in the pretraining stage, which often requires prohibitive computing resources. For this reason, some approaches have avoided the distillation in the pretraining stage and directly perform knowledge distillation in the fine-tuning stage. Unfortunately, fine-tuning distillation will inevitably degrade the performance of model compression because the semantic information acquired by the pretraining stage is largely lost.

To strike an improved tradeoff between cost and performance, we propose to initialize the student model $BERT_6$ using the parameters of the odd-numbered transformer layer in the teacher network. Meanwhile, we propose to distill the even-numbered layer of the teacher network with the student. Specifically, we expect that one module in the student network can at least approximate the reasoning capability of N modules in the teacher network. Toward this objective, we make the head of each module in the student network similar to that of the N modules in the teacher network; and simultaneously make the output of each module in students close to the tail of the N modules in teachers. To keep the student network structure simple, we opt not to add another network module to perform the work of reasoning and restoration. Instead, we will rely on the structure of the feedforward network to restore the inference mode of the

teacher model, which can be expressed as:

$$L_{ffn} = \frac{1}{d} \sum_{i=1}^d MSE(W_i^s, W_{2i}^t) \quad (4)$$

where d is the number of transformer modules in the student network, W_i refers to the parameters of the feedforward network in i -th transformer module, and $MSE()$ denotes the squared error loss function.

Note that the entire process of knowledge distillation training is carried out in the fine-tuning stage only, which avoids the substantial computational cost in the pretraining stage. Inspired by recent work in computer vision [9], the computational cost could be further reduced by partial training. During the knowledge distillation, we only keep the multi-head attention (MHA) module of the student network fixed (i.e., it did not participate in the weight updating). As the parameters involved in retraining are reduced, the total amount of data required by the fine-tuning stage of knowledge distillation can be greatly reduced for specific downstream tasks. By *copying and freezing* half of the teacher network (e.g., odd-numbered layers), its ability can be retained in the pretraining stage. We restore the reasoning mode of the teacher network through the weight constraints of the other half of the feedforward network (e.g., even-numbered layers). Forcing one transformer module in the student network to complete the task of the two transformer modules in the teacher network. This way, we can achieve the objective of the pretrained language model compression with a small computational cost by unifying the fine-tuning stage with the process of knowledge distillation.

Finally, we use a regular Cross Entropy loss for the downstream-specific target tasks:

$$L_{CE}^s = - \sum_{i \in [N]} \sum_{c \in C} [1[y_i = c] \cdot \log P^s(y_i = c|x_i; \Theta^s)] \quad (5)$$

where $x_i \in [N]$ is the i -th training sample; y_i is its corresponding ground-truth label, respectively.

Using the above distillation goals (3), (4), and (5), the objective function for downstream task training of the student network can be reformulated as (α, β are hyperparameters):

$$L_{KD} = L_{CE}^s + \alpha L_{distill} + \beta L_{ffn} \quad (6)$$

where L_{CE}^s represents the specific downstream tasks, $L_{distill}$ represents the distillation of the soft label at the output of the teacher network and the student network, and L_{ffn} represents the weight constraint of the even-numbered layer of the teacher network on the feedforward network (FFN) structure of the student network. Note that the generalization property of the conventional distillation process is usually poor due to the ignorance of feature uncertainty in the latent space [7], [26].

B. Uncertainty-Driven Distillation

Generally speaking, the goal of the classification problem in NLP tasks is to infer its label y from the feature x . Due to the lack of pretraining and natural differences in network structure, it is difficult for the student network to learn the teacher-like feature, especially when the student network is small or the classification problem is difficult.

The reason why uncertainty learning is effective is essential that the noise in the process is modeled by Gaussian distribution, and we believe that there is a certain degree of noise in the process of knowledge distillation. Different from previous studies, we believe that the knowledge imparted by the teacher network is not completely correct. Therefore, by combining knowledge distillation and uncertainty estimation, the idea of modular distillation improves the effect of the proposed method.

We employ finetuned BERT as the teacher network and extract features $\mathbf{z}^t$. The teacher feature $\mathbf{z}^t$ is used as the ground truth for the student backbone network, whose output is $\mathbf{z}^s$.

We pursue a student model capable of performing well by jointly restoring the teacher feature and estimating the class label y simultaneously. By modeling the estimation of y and $\mathbf{z}^t$ as maximum a posterior probability (MAP) estimation, we can use the Bayesian formula to translate the original problem into two subproblems:

$$\arg \max (p(y, \mathbf{z}^t | \mathbf{x})) = \arg \max (p(y | \mathbf{z}^t, \mathbf{x}) p(\mathbf{z}^t | \mathbf{x})) \quad (7)$$

where $p(\mathbf{z}^t | \mathbf{x})$ denotes feature reconstruction and $p(y | \mathbf{z}^t, \mathbf{x})$ denotes a specific NLP task.

We use the deep learning method to address this problem. $h(\cdot | \Theta^h)$ denotes the backbone network in the student model which is expected to be able to reconstruct the teacher feature from $\mathbf{x}$. We model feature uncertainty, which can guide the knowledge transfer in the latent space during training. Assuming that each feature map observes a Gaussian distribution with mean $\mathbf{z}_i^s$ and standard deviation σ_i , we formulate the observation model with the estimated teacher-like feature $\mathbf{z}_i^s$ and target teacher feature $\mathbf{z}_i^t$ as:

$$\mathbf{z}_i^t = \mathbf{z}_i^s + \epsilon \sigma_i \quad (8)$$

where ϵ denotes the normal distribution with zero mean and unit variance. We propose to estimate the mean and uncertainty (variance) σ of teacher-like features $\mathbf{z}_i^s$ simultaneously. To restore the corresponding teacher feature $\mathbf{z}_i^t$, a Gaussian distribution is assumed for representing the likelihood function by:

$$p(\mathbf{z}_i^t | \mathbf{x}_i, \sigma_i) = \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{\|\mathbf{z}_i^t - h(\mathbf{x}_i; \Theta^h)\|^2}{2\sigma_i^2}\right) \quad (9)$$

where $h(\mathbf{x}_i; \Theta^h) = \mathbf{z}_i^s$ denotes the teacher-like feature (mean) and σ_i is the uncertainty (variance). We impose Jeffrey's prior [2]: $p(w) \propto \frac{1}{w}$ on uncertainty estimation σ_i , the above formula can be translated as follows:

$$\begin{aligned} p(\mathbf{z}_i^t, \sigma_i | \mathbf{x}_i) &= p(\mathbf{z}_i^t | \mathbf{x}_i, \sigma_i) p(\sigma_i) \\ &= \frac{1}{\sqrt{2\pi}\sigma_i} \exp\left(-\frac{\|\mathbf{z}_i^t - h(\mathbf{x}_i; \Theta^h)\|^2}{2\sigma_i^2}\right) \frac{1}{\sigma_i} \end{aligned} \quad (10)$$

Then use the log-likelihood function with Jeffrey's prior can be formulated as follows:

$$\ln(\mathbf{z}_i^t, \sigma_i | \mathbf{x}_i) = -\frac{\|\mathbf{z}_i^t - h(\mathbf{x}_i; \Theta^h)\|^2}{2\sigma_i^2} - \ln \sqrt{2\pi} - \ln \sigma_i^2 \quad (11)$$

We add a new branch (UEM) at the end of the student backbone network to estimate the feature uncertainty. Based on the above derivation, we can formulate an uncertainty estimate

distillation loss by maximum likelihood (ML) estimation to help the student network perceive the sample difficulty as:

$$L_{UED} = \frac{1}{n} \sum_{i=1}^n \left(\frac{\|\mathbf{z}_i^t - h(\mathbf{x}_i; \Theta^h)\|^2}{2\sigma_i^2} + \ln \sigma_i^2 \right) \quad (12)$$

where n is the number of samples in a minibatch of the training dataset. We extract teacher features $\mathbf{z}^t$ by the teacher network, and obtain restored teacher-like features $\mathbf{z}^s = h(\mathbf{x}_i; \Theta^h)$ with estimated uncertainty σ through the uncertainty loss function. On the surface, σ represents the difficulty of reconstructing teacher-like features. Deeper reasoning suggests that it might play a role like an attention mechanism to ensure the network focuses on different samples in different stages of training. Therefore, after combining the uncertainty estimation (3), (4), (5), (12), the overall loss function of our scheme is as follows (α, β, γ are hyperparameters):

$$L_{UDKD} = L_{CE}^s + \alpha L_{distill} + \beta L_{ffn} + \gamma L_{UED} \quad (13)$$

where L_{UDKD} denotes the loss of knowledge-driven knowledge distillation and the item L_{UED} and L_{ffn} distinguishes (13) from traditional knowledge distillation.

IV. EXPERIMENTAL RESULTS

In this section, we introduce the experimental details of our Efficient Compression Approach for the BERT Model. It mainly includes the details of the data set and baseline information in Sec. Datasets and Sec. Benchmark Comparisons and further analyze the ablation experiment about uncertainty in Sec. Ablation Study.

A. Datasets

To test the general ability of the language model for the model solution we proposed and other pretrained language model compression solutions, we selected some tasks on the General Language Understanding Evaluation (GLUE) benchmark. The specific information of the five natural language tasks which we have tested is listed as follows.

The Corpus of Linguistic Acceptability (CoLA) is a single-sentence classification task. The corpus comes from books and journals of language theory, and each sentence is marked as a grammatical sequence of words.

The Stanford Sentiment Treebank (SST-2) [27] is a single-sentence classification task. Contains human annotations of sentences in movie reviews and their emotions.

The Microsoft Research Paraphrase Corpus (MRPC) [28] automatically extracts a corpus of sentence pairs from online news sources and manually annotates whether the sentences in the sentence pairs are semantically equivalent.

The Quora Question Pairs (QQP) is a collection of question pairs from the community question and answer website Quora. The task is to determine whether the paired questions are semantically equivalent.

Question-answering NLI dataset (QNLI) [29] is a Natural Language Inference dataset automatically derived from the Stanford Question Answering Dataset v1.1.

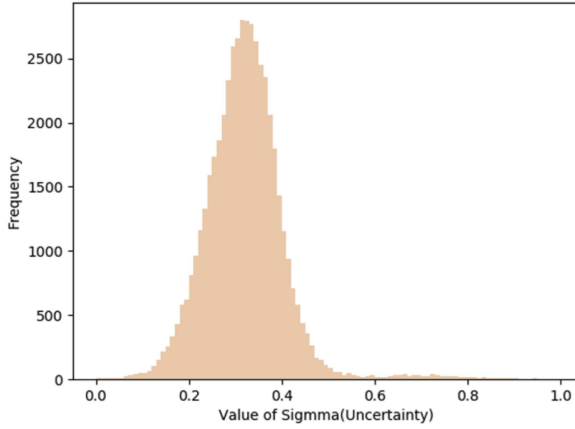


Fig. 2. Distribution of sigma values.

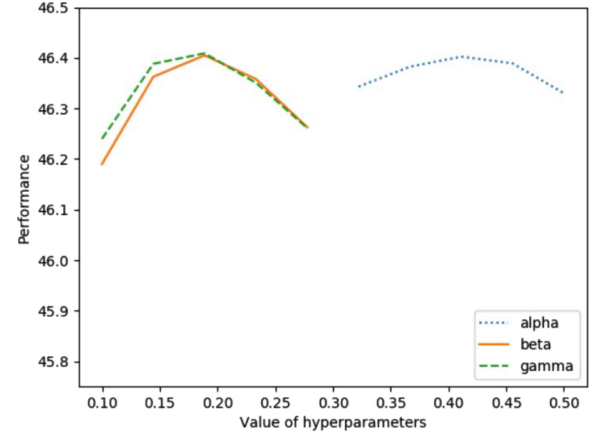


Fig. 3. Parameter analysis.

B. Benchmark Comparisons

The weight of the pretrained language model $BERT_{Base}$ is obtained from the Google release.¹ We fine-tune the $BERT_{Base}$ to specific downstream tasks until the results converge as the teacher network for experiments generally needs 4 to 6 epochs. We initialize a 6-layer transformer module as the student network (denoted as $VanillaBERT_6$), and randomly initialize the parameters in this network. Then we adopt the same measures as the traditional distilling soft label method as BERT vanilla knowledge distillation. The model with the best validation accuracy is selected. In similar research, we chose TheseusBert, PKDBert, and BiLSTM_{SOFT} as strong baselines. For the fairness of comparison, all methods only intervene in the experimental scheme in the fine-tuning stage and do not perform knowledge distillation or other techniques to improve the performance of the student network in the pretraining stage. We have made minimal adjustments to the non-conforming method and ensure the fairness of comparison by preserving the characteristics of the original method.

Meanwhile, we experimented with our scheme when $N = 3$, and the results are shown in Table II. Different compression ratios can be selected for different application scenarios.

Specifically, in our method, we have employed Adam as the optimizer with an initial learning rate of 2×10^{-5} , and our downstream model was trained for about 5 epochs with a batch size of 8. For the hyperparameter in (13), we use $\alpha = 0.4$, $\beta = \gamma = 0.2$ for distinguishing the weight of different loss functions. It is efficient to deploy a model which is suitable for a specific downstream task, without time-consuming pretraining compression (e.g., no longer than 18 GPU hours for each task).

To better explore the role of uncertainty in the knowledge distillation process, we visualized the uncertainty estimation variance of a portion of the input data. The variance distribution of a small batch of data is shown in Fig. 2. The figure shows the uncertainty (σ) distribution learned by the uncertainty estimation module. The uncertainty distribution presents a result similar to the Gaussian distribution, which proves that different data have different uncertainties, that is, different data samples

have different degrees of difficulty. This estimated variance can guide the knowledge distillation training process and assign different training weights to different samples.

C. Ablation Study

We have conducted the General Language Understanding Evaluation (GLUE) [23] benchmark evaluation of our proposed model compression method, which includes a set of different natural language understanding tasks as shown in Table I. Furthermore, we have conducted several ablation studies to investigate which modules significantly contribute to performance improvement as shown in Table III. In our experiments, we roughly divide our method into three modules: feedforward distillation, feature distillation, and uncertainty estimation to verify their impact separately. Feedforward distillation denotes that we use the loss function (6) without uncertainty learning. Feature distillation denotes that we invalidate the uncertainty prediction branch (only use the mean squared error for feature distillation). The last column of the table represents the proposed method in Fig. 1 where we simultaneously learn Teacher-like features (mean) and its uncertainty (variance) loss function in (13).

As in the literature, parameter initialization usually uses the head of the teacher network, and knowledge distillation is applied to the tail of the network. When N is 2, it becomes a copy in the odd-numbered layer and a refinement in even-numbered layers. Table IV shows that the compression performance is slightly better with N modules.

Finally, we conducted an influence degree analysis on the hyperparameters in (13) to explore the influence of different loss terms in the loss function on the results. Fig. 2 shows the corresponding change results of model inference performance when the loss item weight parameters (α , β , γ) in the final loss function change. The results reflect the influence of parameters $\beta > \gamma > \alpha$, which proves the effect of parameter distillation and uncertainty estimation is greater than the loss term of traditional knowledge distillation, which further proves the effectiveness of our proposed method.

The experimental results show that:

¹<http://github.com/google-research/bert>

TABLE I
PERFORMANCE ON DIFFERENT DATASETS

Method	#Params	Interface Time	SST-2	MRPC	CoLA	QQP	QNLI	Average
Bert ₁₂	109M (×1.0)	102s (×1.0)	91.8	89.2	52.1	88.4	90.3	82.36
Bert ₆ —VanillaKD	66M (×1.7)	52.4s (×1.95)	74.8	71.7	34.8	72.3	72.4	65.2
TheseusBert ₆ [Xu <i>et al.</i> , 2020]	67M (×1.6)	52.4s (×1.95)	87.6	86.3	46.3	86.8	85.9	78.58
TheseusBert ₄	53M (×2.1)	36.2s (×2.82)	85.4	83.7	43.5	83.3	84.7	76.12
PKDBert ₆ [Sun <i>et al.</i> , 2019]	67M (×1.6)	52.8s (×1.93)	85.7	81.3	40.2	80.6	85.1	74.58
BiLSTM _{SOFT} [Tang <i>et al.</i> , 2019]	11M (×10.8)	16.5s (×6.2)	84.9	82.0	40.6	82.4	84.4	74.86
Ours₆	55M*/66.8M (×2.2)	53.1s (×1.92)	87.5	86.7	46.8	86.2	86.4	78.72

BERT₁₂ denotes BERT_{base}, PKDBert₆, and TheseusBERT₆ denotes different BERT compression methods which compress BERT into a subscript number layer.
 * Means part of the parameters involved in training (the parameters of the whole model will be a bit more). Since some methods mentioned are not only trained in the fine-tuning stage, but on the premise of keeping the original plan compression idea, we made minimal adjustments to ensure the fairness of the comparison.

TABLE II
EXPERIMENTAL RESULTS OF DIFFERENT COMPRESS RATE N

Method	SST-2	MRPC	CoLA	QQP	QNLI	Avg.
Ours₄	83.2	84.9	41.2	82.7	81.9	74.78
Ours₆	87.5	86.7	46.8	86.2	86.4	78.72

TABLE III
ABLATION STUDY ON OUR MODULES

distill	parm distill	uncertainty	QQP	SST-2	QNLI
√	×	×	76.4	78.8	77.9
×	√	×	83.6	85.2	82.4
√	√	×	82.5	84.2	85.3
√	×	√	84.7	86.0	85.7
√	√	√	86.2	87.5	86.4

TABLE IV
ABOUT THE LOCATION OF THE INITIALIZATION

Method	CoLA	SST-2	MRPC	QQP	QNLI
head and tail	46.4	86.8	86.3	86.1	86.2
Ours(N modules)	46.8	87.5	86.7	86.2	86.4

- 1) The feedforward parameter distillation and uncertainty estimate we proposed are effective for transformer-based model compression. Uncertainty estimation can guide the student network's feature reconstruction in the latent space.
- 2) The model compression method we proposed has less computational burden during training, while the performance is similar to some strong baseline methods, and is more suitable for quickly applying the ability of pretrained language models to a specific task with few sample learning.
- 3) There are many repeating and stacked structures in deep learning. Experiments have proved that there is a possibility of N-to-1 compression for multiple repetitive structures.

D. Limitations and Discussions

We have only considered N-to-1 distillation in this article. Alternative methods of compressing each layer independently

have also shown promising results. Based on previous research experience, combining different compression methods often leads to better performance than the individual counterpart. We have only considered the compression of BERT in this article, and how to explore more complex variations and hybrid of Bi-LSTM, CNN, and transformer models have remained under-researched.

Our method is mainly aimed at uncertainty estimation in knowledge distillation on nature language process tasks (e.g., linguistic acceptability, sentiment classification, semantic equivalence) and has not considered the uncertainty in the internal reasoning process. Since our work improves the performance of the student network in the N-to-1 compression framework through uncertainty estimation and parameter distillation, we can further incorporate uncertainty estimation into block compression to make the combination of module distillation and uncertainty estimate more natural. Future research should be undertaken to explore the modeling uncertainty of the internal module.

V. CONCLUSION

In this article, we propose a new paradigm to compress a pretrained language model by combining uncertainty estimation with knowledge distillation. Our uncertainty-driven knowledge distillation method allows us to achieve a better tradeoff between the conflicting objectives of model compression and accuracy preservation. The original BERT language model can be quickly distilled and compressed to a student model suitable for specific downstream tasks. Experimental results demonstrate the effectiveness of our method on the GLUE benchmark with a smaller model size and faster training time.

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Tianyu Huang received the B.S. degree in science and technology from Hainan University, Hainan, China, in 2019. He is currently working toward the M.S. degree with Xidian University, Xi'an, China. His research interests include natural language processing and deep learning.



Weisheng Dong (Member, IEEE) received the B.S. degree in electronic engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2004, and the Ph.D. degree in circuits and system from Xidian University, Xi'an, China, in 2010. He was a Visiting Student with Microsoft Research Asia, Beijing, China, in 2006. From 2009 to 2010, he was a Research Assistant with the Department of Computing, Hong Kong Polytechnic University, Hong Kong. In 2010, he joined the School of Electronic, Engineering, Xidian University, as a Lecturer,

where he has been a Professor since 2016. His research interests include inverse problems in image processing, deep learning, parse representation. He was the recipient of the Best Paper Award at the SPIE Visual Communication and Image Processing (VCIP) in 2010. He was an Associate Editor for IEEE TRANSACTIONS ON IMAGE PROCESSING and is currently an Associate Editor for *SIAM Journal of Imaging Sciences*.



Fangfang Wu received the B.S. degree in electrical engineering, the M.S. degree in intelligent information processing, and the Ph.D. degree in electronics science and technology from Xidian University, Xi'an, China, in 2008, 2011, and 2021, respectively. In 2022, she joined the School of Computer Science and Technology, Xidian University. Her research interests include image restoration, hyperspectral imaging and deep image representation.



Xin Li (Fellow, IEEE) received the B.S. degree with highest honors in electronic engineering and information science from the University of Science and Technology of China, Hefei, in 1996, and the Ph.D. degree in electrical engineering from Princeton University, Princeton, NJ, USA, in 2000. From 2000 to 2002, he was the Member of Technical Staff with the Sharp Laboratories of America, Camas, WA, USA. Since January 2003, he has been the Faculty Member with the Lane Department of Computer Science and Electrical Engineering. In 2017, he was elected the

Fellow of IEEE for his contributions to image interpolation, restoration, and compression.



Guangming Shi (Fellow, IEEE) received the B.S. degree in automatic control, the M.S. degree in computer control, and the Ph.D. degree in electronic information technology from Xidian University, Xi'an, China, in 1985, 1988, and 2002, respectively. In 1988, he joined the School of Electronic Engineering, Xidian University. From 1994 to 1996, as a Research Assistant, he cooperated with the Department of Electronic Engineering, The University of Hong Kong, Hong Kong. Since 2003, he has been a Professor with the School of Electronic Engineering, Xidian

University. In 2004, he was the Head of National Instruction Base of Electrician and Electronic. From June to December 2004, he studied with the Department of Electronic Engineering, University of Illinois at Urbana-Champaign, Champaign, IL, USA. He is currently the Vice President with Xidian University, and the Academic Leader in the subject of circuits and systems. He has authored or coauthored more than 60 research papers. His research interests include compressed sensing, theory and design of multirate filter banks, image denoising, low-bit-rate image/video coding, and implementation of algorithms for intelligent signal processing (using DSP and FPGA).